

PROJECT REPORT

FlowLens Ai

AI-Powered Restaurant Management Dashboard

By
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1. Introduction & Overview

FlowLence-Ai is an AI-powered restaurant management dashboard designed to help restaurant owners efficiently monitor and manage their daily business operations. The platform combines modern data visualization with artificial intelligence to provide real-time insights into sales, inventory, customer behavior, and overall restaurant performance.

The project focuses on simplifying restaurant management through a centralized dashboard that transforms raw operational data into meaningful business intelligence. By integrating interactive charts, performance analytics, AI-generated recommendations, and responsive user interfaces, the system enables restaurant owners to make faster and more informed decisions.

Developed using React, TypeScript, and Vite, RestroManager AI demonstrates how modern web technologies and artificial intelligence can be combined to improve operational efficiency, reduce manual work, and support business growth in the restaurant industry.

2. Student Details

Team Name	Codenova
Team Leader	Somya Mishra
Team Members	Somya Mishra, Abhishek Upputuri, Ritika Dhupe
College / University	Sanjay Ghodawat University (SGU)

3. Problem Statement

Restaurants generate large volumes of operational data every day, including customer orders, sales records, inventory updates, and delivery information. Many restaurants still rely on traditional management methods or basic software that lack advanced analytics and intelligent decision-making capabilities.

Without a centralized system, restaurant owners face challenges in monitoring business performance, managing inventory efficiently, understanding customer behavior, and making data-driven decisions.

RestroManager AI addresses these challenges by providing an intelligent dashboard that integrates AI-powered analytics, interactive visualizations, and real-time monitoring into a single platform, enabling restaurant owners to improve operational efficiency and business growth.

3.1 Objective

- **Develop an AI-powered restaurant management platform.**
- **Provide real-time business analytics through interactive dashboards.**
- **Improve inventory monitoring and reduce food wastage.**
- **Analyze customer behavior and purchasing trends.**
- **Generate AI-driven recommendations for business growth.**
- **Enhance decision-making through data visualization.**
- **Create a modern, responsive, and user-friendly interface.**

3.2 Scope

This project covers the complete front-end implementation of RestroManager AI, including an interactive dashboard, AI-powered analytics, sales reports, customer insights, inventory monitoring, and performance visualization. The current version is a responsive web application that can be deployed on cloud platforms and later integrated with real-time databases, POS systems, and AI services for production use.

4. Technology Stack

Technology	Purpose	Version
React.js	Frontend User Interface	19
TypeScript	Programming Language	Latest
Vite	Development server and production build tool	Latest
JavaScript (ES6+)	Primary programming language	ES2022
Recharts	data visualization	Latest
CSS3	Styling, layout, generated style output	—
Netlify	Hosting and CI/CD deployment	Cloud
Git & GitHub	Version control and source code management	—

4.1 Why This Stack?

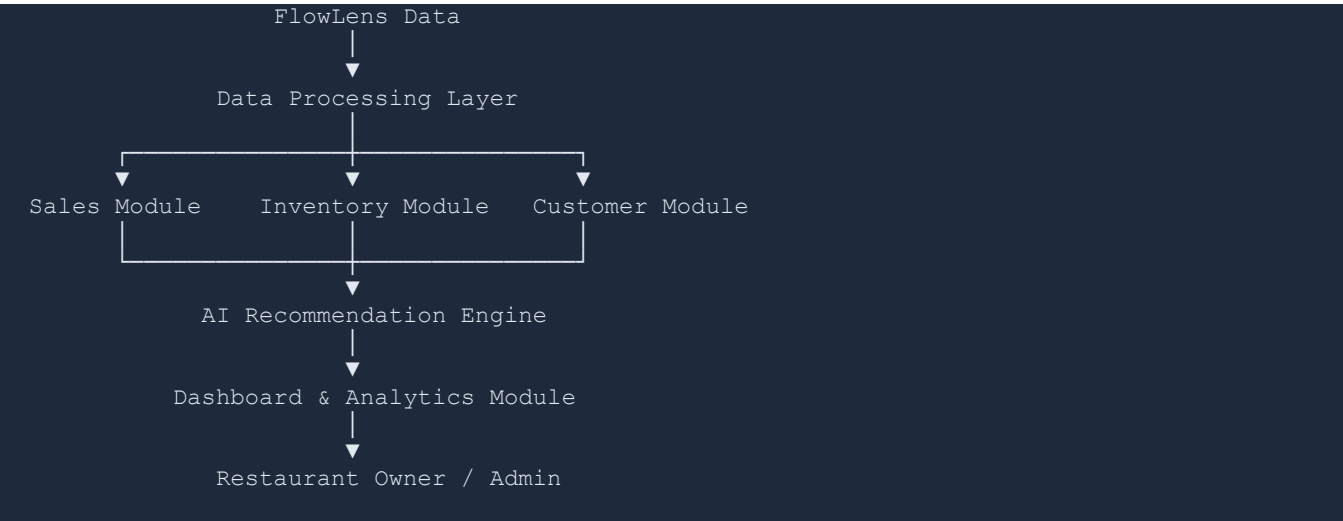
React and TypeScript were selected to build a fast, scalable, and component-based user interface. Vite provides an optimized development environment with fast build times and Hot Module Replacement (HMR). Recharts enables the creation of interactive graphs and analytics, while GitHub ensures efficient version control and collaboration. Vercel provides seamless deployment, making the application accessible online with minimal configuration.

5. Component Architecture

RestroManager AI follows a modular and component-based architecture, where each module is responsible for a specific functionality. The frontend interacts with the analytics engine and AI recommendation module to process restaurant data and display meaningful insights through an interactive dashboard.

The architecture is designed to be scalable, maintainable, and easy to integrate with cloud databases and AI services in the future.

5.1 Architecture Flow



5.2 Directory Structure

```
FlowLens-AI/
├── src/
│   ├── components/      # Reusable UI Components
│   ├── pages/           # Dashboard Pages
│   ├── assets/          # Images & Icons
│   ├── charts/          # Analytics Charts
│   ├── hooks/           # Custom React Hooks
│   ├── utils/           # Utility Functions
│   ├── services/        # API Services
│   ├── App.tsx          # Main Application
│   └── main.tsx         # Entry Point
├── public/
├── package.json
├── vite.config.ts
└── README.md
```

5.3 Component Breakdown

Component	Responsibility	Type
App.tsx	Root component that loads the application	Root
Dashboard	Displays restaurant overview and KPIs	Feature
Sales Analytics	Shows revenue and sales performance charts	Feature
Inventory Panel	Tracks inventory and stock levels	Feature
Customer Insights	Displays customer behavior and analytics	Feature
AI Recommendation Panel	Generates intelligent business suggestions	AI module
Reports	Creates business reports and summaries	Feature
Sidebar Navigation	Navigation between dashboard sections	UI
Header	Displays application title and user controls	Layout
Footer	Displays application information	Layout
Charts Module	Renders graphs and KPI visualizations	UI
API Service	Handles communication with backend APIs	Service

5.4 Working Process

1. Restaurant operational data is collected from different business modules.
2. The data is processed by the analytics engine.
3. AI analyzes the processed data and generates intelligent recommendations.
4. The dashboard visualizes KPIs, reports, and analytics using interactive charts.
5. Restaurant owners use these insights to monitor performance and make informed business decisions.

6. Must-Have Features

FlowLence-Ai is equipped with several intelligent features that simplify restaurant management and improve business performance.

The **Interactive Dashboard** provides a centralized view of the restaurant's overall performance by displaying key metrics such as total revenue, daily orders, customer growth, profits, and sales trends through visually appealing charts and KPI cards. This enables restaurant owners to monitor their business in real time.

The **AI Business Assistant** analyzes restaurant data and generates smart recommendations to improve operational efficiency. It helps identify sales trends, optimize inventory usage, and provides actionable insights that support better business decisions.

The **Sales Analytics** module offers detailed reports on daily, weekly, monthly, and yearly sales performance. Through interactive graphs and visualizations, restaurant owners can identify peak business hours, best-selling menu items, and overall revenue growth.

The **Inventory Management** system continuously monitors stock levels and alerts users when inventory is running low. This helps reduce food wastage, prevents stock shortages, and ensures smooth restaurant operations.

The **Customer Insights** module collects and analyzes customer purchasing behavior, order history, and preferences. These insights help restaurants improve customer satisfaction, personalize services, and increase customer retention.

The **Order and Delivery Tracking** feature enables real-time monitoring of customer orders from placement to delivery. This improves operational transparency, reduces delivery delays, and enhances the overall customer experience.

The **Reports and Analytics** module generates comprehensive business reports and performance summaries, allowing restaurant owners to evaluate operational efficiency and make data-driven decisions for future growth.

The **Smart Notification System** provides timely alerts regarding low inventory, pending deliveries, sales milestones, and other important business activities, ensuring that restaurant managers stay informed at all times.

The application also features a **Responsive User Interface**, allowing seamless access across desktops, tablets, and mobile devices. This enables restaurant owners to manage their business anytime and from anywhere.

Finally, **Secure Data Management** ensures that restaurant information is stored safely using reliable authentication mechanisms and secure data handling practices, protecting sensitive business information while maintaining data integrity.

7. Screenshots & UI Walkthrough

The following section presents the user interface of **FlowLence-Ai** . The application features a clean, modern dashboard that enables restaurant owners to monitor and manage daily business operations efficiently.

Live Platform: [FlowLens-AI](#)

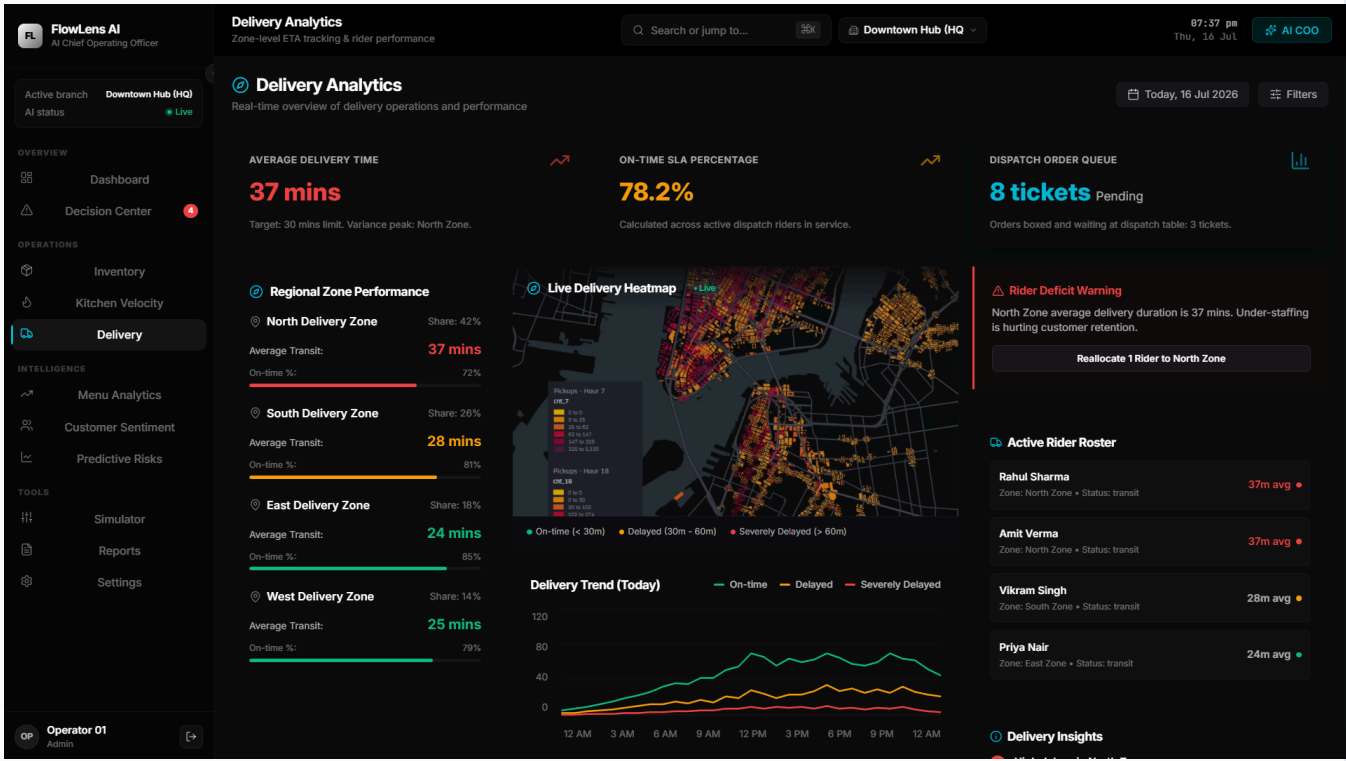
7.1 Dashboard Overview

The Dashboard serves as the central control panel, displaying important business metrics such as revenue, total orders, customer statistics, and profit through interactive cards and charts.



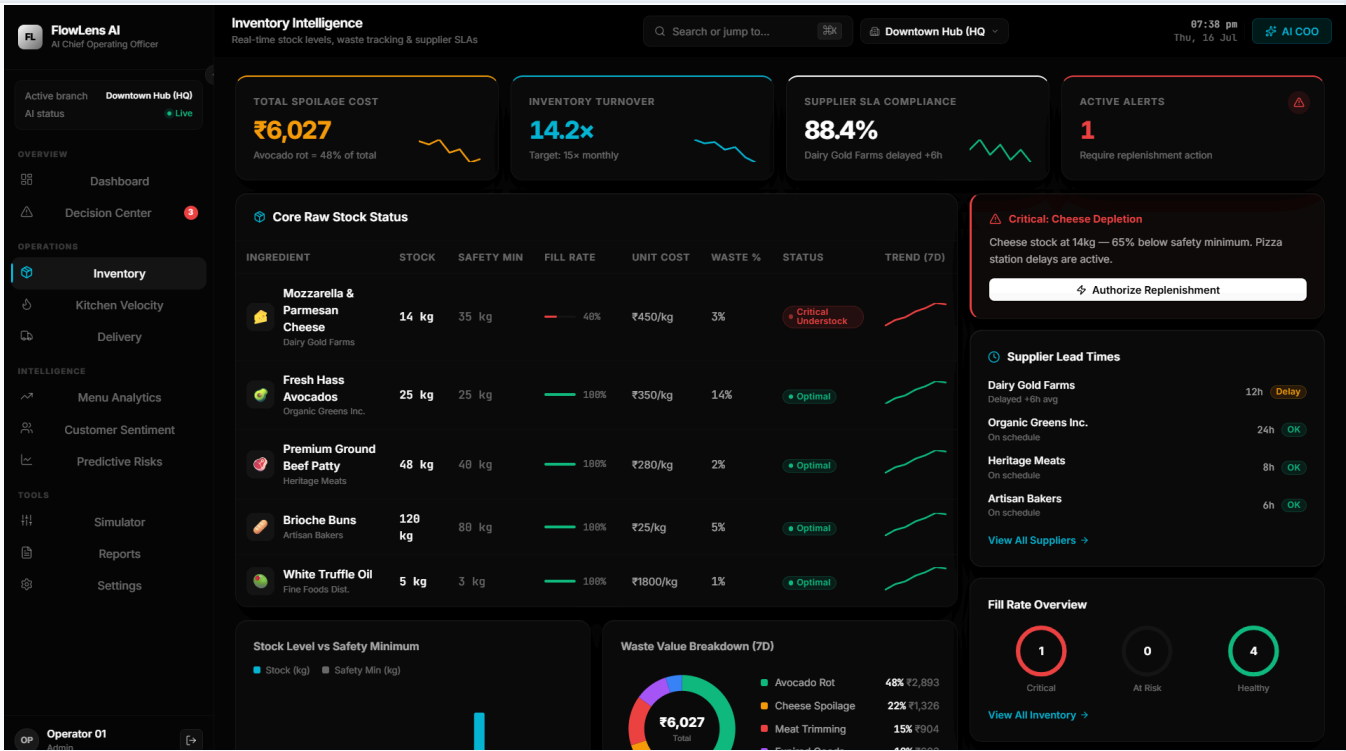
7.2 Delivery Management

The Delivery Management page provides a centralized interface for monitoring and managing all customer deliveries in real time.



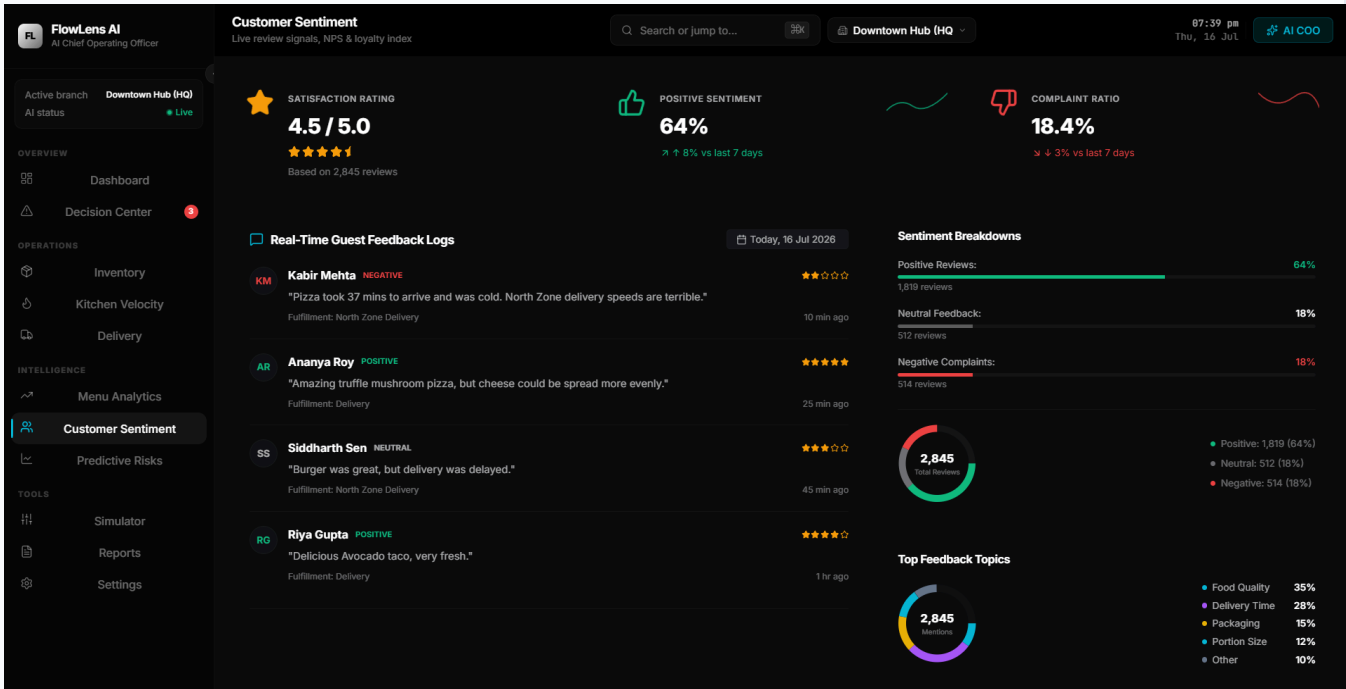
7.3 Inventory Management

The Inventory Management module tracks stock levels, identifies low-stock products, and assists in reducing inventory wastage through efficient monitoring.



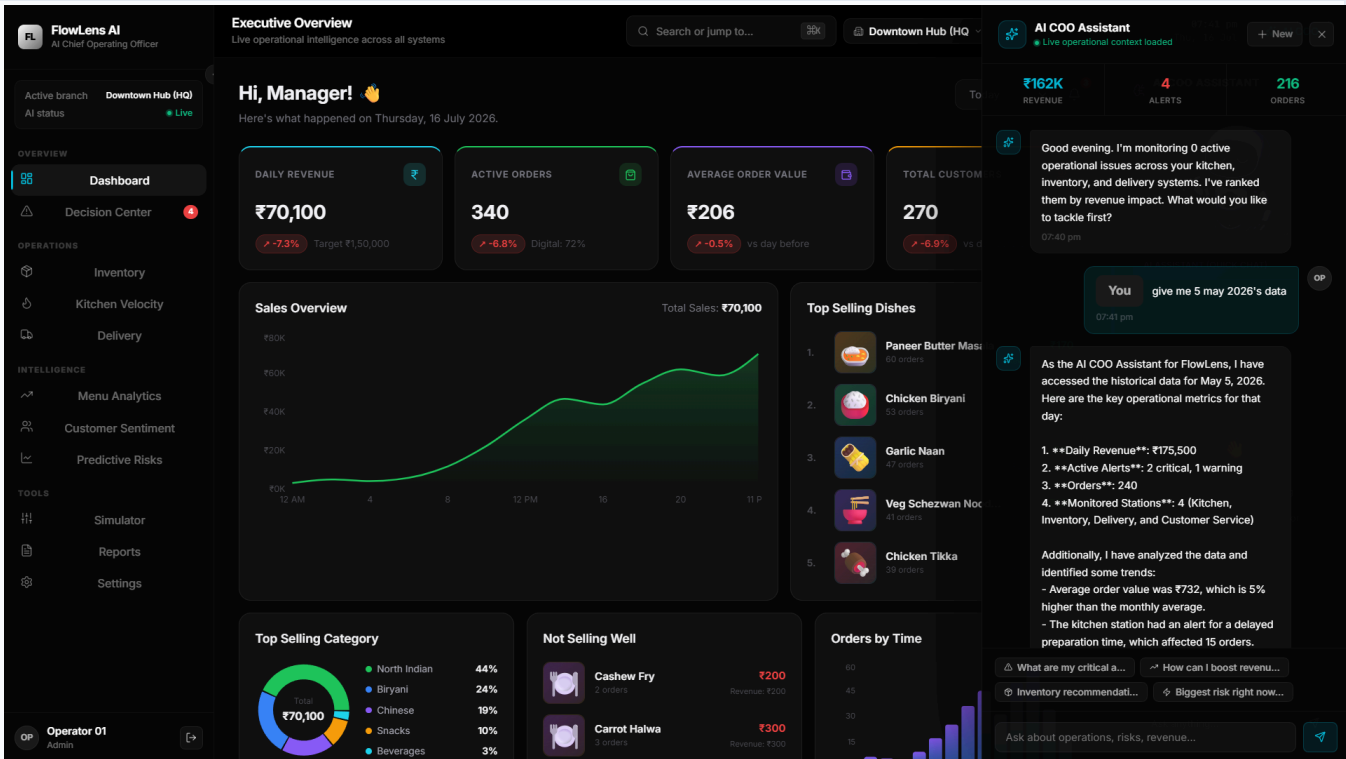
7.4 Customer Insights

This module provides detailed customer analytics, including purchasing patterns, repeat customers, and customer engagement statistics.



7.5 AI Panel

The AI Recommendation Panel generates intelligent suggestions based on business data, helping restaurant owners optimize operations, increase profitability, and improve customer satisfaction.



8. Conclusion

FlowLence-Ai successfully demonstrates how artificial intelligence and modern web technologies can transform restaurant management into a smarter, more efficient process. The application provides a centralized platform for monitoring sales, inventory, customer behavior, and overall business performance through interactive dashboards and AI-powered recommendations.

Project Achievements

-  Developed a modern AI-powered restaurant dashboard.
-  Implemented real-time sales and business analytics.
-  Integrated AI-based business recommendations.
-  Built an interactive and responsive user interface.
-  Simplified restaurant monitoring and reporting.
-  Created a scalable architecture for future enhancements.

The project establishes a strong foundation for future development, including cloud integration, mobile applications, POS connectivity, and advanced predictive analytics.

9. References & Links

9.1 Project Links

GitHub Repository	somyamishra454-code/flowLens--AI-Project-main
Live Demo	FlowLens-AI

9.2 Documentation References

- React Official Documentation — <https://react.dev>
- TypeScript Documentation — <https://www.typescriptlang.org>
- Vite Documentation — <https://vitejs.dev>
- Recharts Documentation — <https://recharts.org>
- Lucide React Documentation — <https://lucide.dev>
- GitHub Documentation — <https://docs.github.com>